Matrices.

Definition of matrix:

A rectangular arrangement of numbers (which may be real or complex numbers) in rows and columns, is called a matrix. This arrangement is enclosed by small ( ) or big [ ] brackets. The numbers are called the elements of the matrix or entries in the matrix.

Order of a matrix:

A matrix having ***m* rows** and ***n* columns** is called a matrix of order $\boldsymbol{m \times n}$ matrix (read as an $m\ by\ n$ matrix).

$$A = \begin{bmatrix} -1 & 2 & -3 \\ 4 & 0 & 2 \end{bmatrix}\ of\ order\ 2 \times 3. \quad , \quad B = \begin{bmatrix} 2 & 3 \\ 0 & 5 \\ 1 & 4 \end{bmatrix}\ of\ order\ 3 \times 2$$

$$C = \begin{bmatrix} 3 & 2 \\ 1 & 5 \end{bmatrix}\ of\ order\ 2 \times 2 \quad , \quad D = \begin{bmatrix} 2 & -1 & 0 & 5 \end{bmatrix}\ of\ order\ 1 \times 4$$

Types of Matrices:

1. ***Rectangular matrix:*** *:* A matrix having m rows and n columns is $(of\ order\ \boldsymbol{m \times n})$ such that $\boldsymbol{m \neq n}$.

2. ***Square matrix:*** A matrix in which the number of rows equals its number of columns.

3. ***Row matrix:*** A matrix has only one row and any number of columns.

$$A_{1\times3} = \begin{bmatrix} 2 & 4 & 1 \end{bmatrix}\ row\ matrix\ of\ order\ 1 \times 3. \quad \& \quad B_{1\times5} = \begin{bmatrix} 3 & 0 & 1 & 5 & 7 \end{bmatrix}$$

4. ***Column matrix:*** A matrix has only one column and any number of rows.

$$C_{3\times1} = \begin{bmatrix} 2 \\ -1 \\ 4 \end{bmatrix}\ coulmn\ matrix\ of\ order\ 3 \times 1 \quad \& \quad C_{4\times1} = \begin{bmatrix} 3 \\ 0 \\ 2 \\ 5 \end{bmatrix}\ coulmn\ matrix\ of\ order\ 4 \times 1$$

5. ***Diagonal matrix:*** A square matrix has only elements on its diagonal and all other elements are zeros.

$$C = \begin{bmatrix} 2 & 0 \\ 0 & 5 \end{bmatrix}\ diagonal\ matrix\ of\ order\ 2 \times 2 \quad D = \begin{bmatrix} 3 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 1 \end{bmatrix}\ diagonal\ matrix\ of\ order\ 3 \times 3$$

6. ***Singleton matrix:*** A matrix has only one none zero element.

7. ***Null or Zero matrix:*** A matrix in which all the elements are zeros.

$$O_1 = [0] \quad , \quad O_2 = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \quad , \quad O_3 = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \quad , \quad O_{3\times2} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \end{bmatrix} \quad , \quad O_{2\times3} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

8. ***Scalar matrix****:* A square matrix whose all non-diagonal elements are zero and diagonal elements are equal.

9. ***Identity matrix:*** A square matrix in which the all diagonal elements are 1 and the rest are all zeros.

$$I_1 = [1] \quad , \quad I_2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \quad , \quad I_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

10. ***Triangular matrix:*** A square matrix in which each element above or below the principal diagonal is zero.

$$A = \begin{bmatrix} 2 & 1 & 3 \\ 0 & 0 & 4 \\ 0 & 0 & 5 \end{bmatrix} is\ upper\ triangular\ matrix, \quad and \quad B = \begin{bmatrix} 3 & 0 & 0 \\ 1 & 0 & 0 \\ 2 & 6 & 7 \end{bmatrix}\ is\ lower\ triangular\ matrix$$

Equality of Matrices

Two matrices A and B are said to be equal matrix if they are of same order and their corresponding elements are equal.

Addition and subtraction:

Order of the matrices must be the same, then we add (subtract) corresponding elements together.

i) $If\ \ A = \begin{bmatrix} -1 & 2 & -3 \\ 4 & 0 & 2 \end{bmatrix}\ \ and\ \ B = \begin{bmatrix} 2 & 0 & 5 \\ 3 & 1 & 3 \end{bmatrix}, then\ \ \ \boldsymbol{A+B} = \begin{bmatrix} \mathbf{1} & \mathbf{2} & \mathbf{2} \\ \mathbf{7} & \mathbf{1} & \mathbf{5} \end{bmatrix}$

ii) $If\ \ C = \begin{bmatrix} 2 & 3 \\ 0 & 5 \\ 1 & 4 \end{bmatrix}\ \ and\ \ D = \begin{bmatrix} 1 & 0 \\ 2 & 3 \\ 5 & 1 \end{bmatrix}$ *then,* $\boldsymbol{C+D} = \begin{bmatrix} \mathbf{2} & \mathbf{3} \\ \mathbf{0} & \mathbf{5} \\ \mathbf{1} & \mathbf{4} \end{bmatrix} + \begin{bmatrix} \mathbf{1} & \mathbf{0} \\ \mathbf{2} & \mathbf{3} \\ \mathbf{5} & \mathbf{1} \end{bmatrix} = \begin{bmatrix} \mathbf{3} & \mathbf{3} \\ \mathbf{2} & \mathbf{8} \\ \mathbf{6} & \mathbf{5} \end{bmatrix}$ *and*

$$\boldsymbol{C-D} = \begin{bmatrix} \mathbf{2} & \mathbf{3} \\ \mathbf{0} & \mathbf{5} \\ \mathbf{1} & \mathbf{4} \end{bmatrix} - \begin{bmatrix} \mathbf{1} & \mathbf{0} \\ \mathbf{2} & \mathbf{3} \\ \mathbf{5} & \mathbf{1} \end{bmatrix} = \begin{bmatrix} \mathbf{1} & \mathbf{3} \\ \mathbf{-2} & \mathbf{2} \\ \mathbf{-4} & \mathbf{3} \end{bmatrix}$$

Properties of Matrix Addition:

Let $A, B, and\ C$ be $\mathrm{m} \times \mathrm{n}$ matrices and $r, s \in R$

1. $A + B = B + A$ (Matrix addition is commutative.)

2. $A + (B + C)\ =\ (A + B) + C$ (Matrix addition is associative.)

3. $r(A + B) =\ rA + rB$ (Scalar multiplication distributes over matrix addition.)

4. $(r + s)A\ =\ rA + sA$ (Real number addition distributes over scalar multiplication.)

5. $(rs)A\ =\ r(sA)$ (An associativity for scalar multiplication.)

6. There is a unique $m \times\ n$ matrix O such that for any $m \times n$ matrix M, $M + O = M$.

(This is called the $m \times n$ zero matrix.)

7. For every $m \times n$ matrix M there is a unique m × n matrix N such that $M + N\ =\ O$

(This N is called the negative of M and is denoted −M.)

Matrix Multiplication: $\boldsymbol{A_{m\times n} \times B_{n\times p} = C_{m\times p}}$

The number of columns in the first matrix **must be** equal to the number of rows in the second matrix and the order of the product is the number of rows in the first matrix by the number of columns in the second matrix.

i) $If\ \ A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}\ \ and\ \ B = \begin{bmatrix} 1 & -2 & 3 \\ 4 & 0 & 2 \end{bmatrix}\ then,\ \ AB = \begin{bmatrix} 9 & -2 & 7 \\ 19 & -6 & 17 \end{bmatrix}$ and,

ii) $If\ \ \ C = \begin{bmatrix} -1 & 2 & -3 \\ 4 & 0 & 2 \end{bmatrix}\ \ and\ \ D = \begin{bmatrix} 2 & 3 \\ 0 & 5 \\ 1 & 4 \end{bmatrix}\ \ then,$

$$CD = \begin{bmatrix} -1 & 2 & -3 \\ 4 & 0 & 2 \end{bmatrix} \begin{bmatrix} 2 & 3 \\ 0 & 5 \\ 1 & 4 \end{bmatrix} = \begin{bmatrix} -5 & -5 \\ 10 & 20 \end{bmatrix}\ \ and\ \ DC = \begin{bmatrix} 2 & 3 \\ 0 & 5 \\ 1 & 4 \end{bmatrix} \begin{bmatrix} -1 & 2 & -3 \\ 4 & 0 & 2 \end{bmatrix} = \begin{bmatrix} 10 & 4 & 0 \\ 20 & 0 & 10 \\ 15 & 2 & 5 \end{bmatrix}$$

Properties of Matrix Multiplication:

Let A, B, and C be matrices of the appropriate sizes and $r \in R$, then

1. $\boldsymbol{A(BC) = (AB)C}$ (Matrix multiplication is associative.)

2. $\boldsymbol{(rA)B = r(AB) = A(rB)}$ (Scalar multiplication commutes with matrix multiplication.)

3. $\boldsymbol{A(B + C) = AB + AC.}$

4. $\boldsymbol{(A + B)C = AC + BC}$. (Matrix multiplication distributes over matrix addition.)

5. There exists a unique $n \times n$ matrix I such that for all $n \times n$ matrices M, $IM = MI = M$.

   (I *is called the* $n \times n$ *identity* matrix.)

Remarks on Matrix Multiplication

i) $In\ general,\ \boldsymbol{AB \neq BA}\quad (i.e, Matrix\ multiplication\ is\ not\ commutative).$

$$If\ \ A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \quad and \quad B = \begin{bmatrix} 1 & 1 \\ 0 & 2 \end{bmatrix} \;\Longrightarrow\; AB = \begin{bmatrix} 1 & 5 \\ 3 & 11 \end{bmatrix} \quad and \quad BA = \begin{bmatrix} 4 & 6 \\ 6 & 8 \end{bmatrix}$$

ii) If A and B $are\ square\ matrices\ such\ that$ $\mathbf{AB = BA}$, $then$ A and B $are\ said\ to$ $\mathbf{commute}$.

$$If\ \ C = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix} \quad and \quad D = \begin{bmatrix} 3 & 0 \\ 0 & 4 \end{bmatrix} \qquad \Longrightarrow \qquad CD = DC = \begin{bmatrix} 3 & 0 \\ 0 & 8 \end{bmatrix}$$

iii) $If\ A = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}\ ,\ B = \begin{bmatrix} 2 & 1 \\ 3 & 2 \end{bmatrix}\ ,\ C = \begin{bmatrix} -2 & 7 \\ 5 & -1 \end{bmatrix}$

$$AB = AC = \begin{bmatrix} 8 & 5 \\ 16 & 10 \end{bmatrix} \qquad but \qquad \boldsymbol{B \neq C} \qquad !!!!!!!!!!!!!!!!!!!!!!$$

iv) $Let\ \ A = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}\ ,\ B = \begin{bmatrix} 4 & -6 \\ -2 & 3 \end{bmatrix} \qquad AB = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}\ \ but\ \ \boldsymbol{A \neq O\ or\ \ B \neq O} \qquad !!!!!!!!!!!!$

$$A = \begin{bmatrix} 4 & 2 \\ -8 & -4 \end{bmatrix} \quad and \quad A^2 = A.A = \begin{bmatrix} 4 & 2 \\ -8 & 4 \end{bmatrix}\begin{bmatrix} 4 & 2 \\ -8 & 4 \end{bmatrix} = \begin{bmatrix} \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} \end{bmatrix} \quad \boldsymbol{but}\ \ \boldsymbol{A \neq O} \qquad !!!!!!!!!!!!$$

Transpose of a Matrix:

If $\boldsymbol{A}$ is any $\boldsymbol{m} \times \boldsymbol{n}$ matrix, then the ***transpose of A***, denoted by $\boldsymbol{A^T}$ and is defined to be the $\boldsymbol{n} \times \boldsymbol{m}$ matrix that results from interchanging the rows and columns of $\boldsymbol{A}$; that is, the first column of $\boldsymbol{A^T}$ is the first row of $\boldsymbol{A}$, the second column of $\boldsymbol{A^T}$ is the second row of $\boldsymbol{A}$, and so forth.

$$\boldsymbol{If\ \ A = \begin{bmatrix} -1 & 2 & -3 \\ 4 & 0 & 2 \end{bmatrix},\ \ then \qquad A^T = \begin{bmatrix} -1 & 4 \\ 2 & 0 \\ -3 & 2 \end{bmatrix}}\ \textbf{and,}$$

$$If\ B = \begin{bmatrix} 2 & 5 & 4 \\ 3 & 1 & 2 \\ 5 & 4 & 6 \end{bmatrix},\ then \quad B^T = \begin{bmatrix} 2 & 3 & 5 \\ 5 & 1 & 4 \\ 4 & 2 & 6 \end{bmatrix}$$

Properties of Transpose:

If A *and* B are matrices and k is scalar, then

1. $(AB)^T = B^T A^T$
2. $(A \pm B)^T = A^T \pm B^T$
3. $(A^T)^T = A$
4. $(kA)^T = k.A^T$
5. $|A^T| = |A|$
6. $(A^T)^{-1} = (A^{-1})^T$

Trace of a matrix:

If A is a square matrix, then the *trace of A*, denoted by $tr(A)$ and is defined to be the sum of the entries on the main diagonal of A. The trace of A is undefined if A is not a square matrix.

Example: $If\ B = \begin{bmatrix} 2 & 5 & 4 \\ 3 & 1 & 2 \\ 5 & 4 & 6 \end{bmatrix}\ then, \quad tr(A) = 2 + 1 + 6 = 9$

Properties of the trace:

For any square matrix A , B and any scalar k

1. $tr(A + B) = tr(A) + tr(B)$.
2. $tr(kA) = k.tr(A)$.
3. $tr(AB) = tr(BA)$
4. $tr(A) = tr(A^T)$

Determinant:

It is a square array of numbers or variables enclosed between parallel vertical bars.

To find a determinant of matrix you must have a ***SQUARE MATRIX.***

We know that if A is a square matrix, then its determinant is denoted by $det(A)$ or $|A|$.

If $A = \begin{bmatrix} 2 & 5 & 4 \\ 3 & 1 & 2 \\ 5 & 4 & 6 \end{bmatrix}$, then $|A| = \begin{vmatrix} 2 & 5 & 4 \\ 3 & 1 & 2 \\ 5 & 4 & 6 \end{vmatrix} = 2\begin{vmatrix} 1 & 2 \\ 4 & 6 \end{vmatrix} - 5\begin{vmatrix} 3 & 2 \\ 6 & 5 \end{vmatrix} + 4\begin{vmatrix} 3 & 1 \\ 5 & 4 \end{vmatrix} = 2(-2) - 5(8) + 4(7) = -16$

Properties of Determinants:

1) Let A be $n \times n$ matrix, then
   - i) If A has a row or column consisting entirely of zeros, then $det(A) = 0$.
   - ii) If A has two identical rows or two identical columns, then $det(A) = 0$.
2) If A and B are $n \times n$ matrices, then $det(AB) = det(A).det(B)$ **Or,** $|AB| = |A||B|$
3) If A is $n \times n$ matrix, then $det(\alpha A) = \alpha^n det(A)$**. Or,** $|\alpha A| = \alpha^n |A|$
4) If A is $n \times n$ matrix, then $det(A^T) = det(A)$**. Or,** $|A^T| = |A|$
5) If A is a ***diagonal matrix***, then $det(A)$ equals the product of the diagonal elements of A.

The Inverse of a matrix:

If $\boldsymbol{A}$ is a square matrix, and if a matrix $\boldsymbol{B}$ of the same size can be found such that $\boldsymbol{AB} = \boldsymbol{BA} = \boldsymbol{I}$, then $\boldsymbol{A}$ is said to be *invertible* and $\boldsymbol{B}$ is called an ***inverse*** of $\boldsymbol{A}$. If no such matrix $\boldsymbol{B}$ can be found, then $\boldsymbol{A}$ is said to be *singular*.

Definition (1): For any *singular matrix* $\boldsymbol{A}$, $\quad |\boldsymbol{A}| = \boldsymbol{0}$

Definition (2): For any non-singular matrix $\boldsymbol{A}$ there is an inverse (unique) denoted by $\boldsymbol{A}^{-1}$ such that; $\boldsymbol{A}.\boldsymbol{A}^{-1} = \boldsymbol{A}^{-1}.\boldsymbol{A} = \boldsymbol{I}$ and is given by the formula:

$$\boldsymbol{A}^{-1} = \frac{\boldsymbol{1}}{\boldsymbol{det(A)}}.\boldsymbol{adj(A)}$$

Example: If $A = \begin{bmatrix} 2 & 1 & 2 \\ 3 & 2 & 2 \\ 1 & 2 & 3 \end{bmatrix}$, *compute* $\boldsymbol{adj(A)}$ *and* $\boldsymbol{A}^{-1}$.

Solution

$$\boldsymbol{adj(A)} = \begin{bmatrix} +\begin{vmatrix} 2 & 2 \\ 2 & 3 \end{vmatrix} & -\begin{vmatrix} 3 & 2 \\ 1 & 3 \end{vmatrix} & +\begin{vmatrix} 3 & 2 \\ 1 & 2 \end{vmatrix} \\ -\begin{vmatrix} 1 & 2 \\ 2 & 3 \end{vmatrix} & +\begin{vmatrix} 2 & 2 \\ 1 & 3 \end{vmatrix} & -\begin{vmatrix} 2 & 1 \\ 1 & 2 \end{vmatrix} \\ +\begin{vmatrix} 1 & 2 \\ 2 & 2 \end{vmatrix} & -\begin{vmatrix} 2 & 2 \\ 3 & 2 \end{vmatrix} & +\begin{vmatrix} 2 & 1 \\ 3 & 2 \end{vmatrix} \end{bmatrix}^{\boldsymbol{T}} = \begin{bmatrix} 2 & -7 & 4 \\ 1 & 4 & -3 \\ -2 & 2 & 1 \end{bmatrix}^{T} = \begin{bmatrix} 2 & 1 & -2 \\ -7 & 4 & 2 \\ 4 & -3 & 1 \end{bmatrix}$$

$$\boldsymbol{det(A)} = \begin{vmatrix} 2 & 1 & 2 \\ 3 & 2 & 2 \\ 1 & 2 & 3 \end{vmatrix} = 2\begin{vmatrix} 2 & 2 \\ 2 & 3 \end{vmatrix} - 1\begin{vmatrix} 3 & 2 \\ 1 & 3 \end{vmatrix} + 2\begin{vmatrix} 3 & 2 \\ 1 & 2 \end{vmatrix} = 2(2) - 1(7) + 2(4) = 4 - 7 + 8 = 5$$

Therefore, $\quad \boldsymbol{A}^{-1} = \frac{\boldsymbol{1}}{\boldsymbol{det(A)}}\boldsymbol{adj(A)} = \frac{1}{5}\begin{bmatrix} 2 & 1 & -2 \\ -7 & 4 & 2 \\ 4 & -3 & 1 \end{bmatrix}$

Remark: **Only** for any 2×2 *invertible (nonsingular)* matrix, $\boldsymbol{A} = \begin{bmatrix} \boldsymbol{a} & \boldsymbol{b} \\ \boldsymbol{c} & \boldsymbol{d} \end{bmatrix}$, then; $\quad \boldsymbol{A}^{-1} = \frac{\boldsymbol{1}}{\boldsymbol{det(A)}}\begin{bmatrix} \boldsymbol{d} & -\boldsymbol{b} \\ -\boldsymbol{c} & \boldsymbol{a} \end{bmatrix}$

Example: *If* $\boldsymbol{A} = \begin{bmatrix} \boldsymbol{3} & \boldsymbol{2} \\ \boldsymbol{5} & \boldsymbol{4} \end{bmatrix}$, then; $\quad \boldsymbol{A}^{-1} = \frac{\boldsymbol{1}}{|\boldsymbol{A}|}\begin{bmatrix} \boldsymbol{4} & -\boldsymbol{2} \\ -\boldsymbol{5} & \boldsymbol{3} \end{bmatrix} = \frac{1}{2}\begin{bmatrix} \boldsymbol{4} & -\boldsymbol{2} \\ -\boldsymbol{5} & \boldsymbol{3} \end{bmatrix}$

Properties of Matrix inverse:

Let $\boldsymbol{A}$ *and* $\boldsymbol{B}$ are non-singular (invertible) matrices, then

1. ***$\boldsymbol{A}^{-1}$ is unique.***
2. $(\boldsymbol{AB})^{-1} = \boldsymbol{B}^{-1}\boldsymbol{A}^{-1}$ and $(\boldsymbol{A} + \boldsymbol{B})^{-1} \neq \boldsymbol{A}^{-1} + \boldsymbol{B}^{-1}$.
3. $(\boldsymbol{A}^{-1})^{-1} = \boldsymbol{A}$.
4. $(\boldsymbol{A}^{k})^{-1} = (\boldsymbol{A}^{-1})^{k}$
5. $(\boldsymbol{kA})^{-1} = \frac{\boldsymbol{1}}{\boldsymbol{k}}\boldsymbol{A}^{-1}$

Some special matrices

Orthogonal matrix

A square matrix A is called *orthogonal matrix* if, $AA^T = A^T A = I$, *where I is the identity matrix*.
In other words, a matrix A is called orthogonal if its transpose is equal to its inverse: $A^T = A^{-1}$.

Remark: For any orthogonal matrix , $\det(A) = \pm 1$.

A matrix, $A = \begin{bmatrix} \cos x & \sin x \\ -\sin x & \cos x \end{bmatrix}$ *is orthogonal because,* $AA^T = A^T A = I$ and $\det(A) = 1$.

In general, there is an infinite family of orthogonal matrices of the form: $\begin{bmatrix} a & c & b \\ b & a & c \\ c & b & a \end{bmatrix}$

Symmetric Matrix

A *square matrix* A is called a *symmetric matrix* if, $A^T = A$ .The entries of a *symmetric matrix* are *symmetric* with respect to the main diagonal.

$$A = \begin{bmatrix} 2 & 5 \\ 5 & 4 \end{bmatrix} \qquad \& \qquad B = \begin{bmatrix} 1 & 4 & 5 \\ 4 & 2 & -6 \\ 5 & -6 & 3 \end{bmatrix} \quad \textit{are symmetric matrices.}$$

Skew Symmetric Matrix

A *square matrix* A is called a *skew symmetric matrix* if, $A^T = -A$ .The entries of a *symmetric matrix* are *symmetric* with respect to the main diagonal with different sign.

$$C = \begin{bmatrix} 0 & -5 \\ 5 & 0 \end{bmatrix} \qquad \& \qquad D = \begin{bmatrix} 0 & 4 & -5 \\ -4 & 0 & 6 \\ 5 & -6 & 0 \end{bmatrix} \quad \textit{are skew symmetric matrices.}$$

Remark: *In any skew symmetric matrix, all diagonal elements are zeros.*

Periodic Matrix

A square matrix A is called *periodic matrix of period n* if there is a positive integer n (the least integer) such that, $A^{n+1} = A$, or in other words, if $A^n = I$

The matrix, $A = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$ *is periodic matrix of period* ***3*** *because*, $A^3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = I$

Remark: In fact if A is periodic matrix of period n, then $A^n = \begin{cases} A & \text{if } n \text{ is odd} \\ I & \text{if } n \text{ is even} \end{cases}$

Definition: If $A^2 = I$, then the periodic matrix A is called ***Involutory matrix***. In other words, if, $A^{-1} = A$.

For example,

i) $\boldsymbol{A} = \begin{pmatrix} \mathbf{1} & \mathbf{0} \\ \mathbf{-1} & \mathbf{-1} \end{pmatrix}$ *is an involutory matrix because,* $\boldsymbol{A^2 = I}$ **or** $\boldsymbol{A = A^{-1}}$.

ii) $\boldsymbol{B} = \begin{pmatrix} \mathbf{1} & \mathbf{0} \\ \mathbf{1} & \mathbf{-1} \end{pmatrix}$ *is an involutory matrix because,* $\boldsymbol{A^2 = I}$ **or** $\boldsymbol{B = B^{-1}}$.

iii) $\boldsymbol{C} = \begin{pmatrix} \mathbf{3} & \mathbf{2} \\ \mathbf{-4} & \mathbf{-3} \end{pmatrix}$ *is an involutory matrix because,* $\boldsymbol{A^2 = I}$ **or** $\boldsymbol{C = C^{-1}}$.

Idempotent Matrix

A square matrix $\boldsymbol{A}$ is called *idempotent matrix* if, $\boldsymbol{A^2 = A}$.

For example,

$\boldsymbol{I} = \begin{pmatrix} \mathbf{1} & \mathbf{0} \\ \mathbf{0} & \mathbf{1} \end{pmatrix}$, *is* (2×2) *idempotent matrix because,* $\boldsymbol{I^2 = I}$ and

$$\boldsymbol{A} = \begin{bmatrix} \mathbf{2} & \mathbf{-2} & \mathbf{-4} \\ \mathbf{-1} & \mathbf{3} & \mathbf{4} \\ \mathbf{1} & \mathbf{-2} & \mathbf{-3} \end{bmatrix} \text{ is } (3 \times 3) \text{ idempotent matrix because, } \boldsymbol{A^2 = A}$$

Remark: *With the exception of the identity matrix, an idempotent matrix is singular.*

Problems:

i) If $\boldsymbol{K}$ is an idempotent matrix, show that $\boldsymbol{I} - \boldsymbol{K}$ is an idempotent.

ii) Suppose that $\boldsymbol{A}$ *and* $\boldsymbol{B}$ are idempotent matrices such that $\boldsymbol{A} + \boldsymbol{B}$ is idempotent, prove that $\boldsymbol{AB} = \boldsymbol{BA} = 0$.

Nilpotent Matrix:

A square matrix $\boldsymbol{A}$ is called *nilpotent matrix* if and only if, $\boldsymbol{A^n = O}$, for a positive integer $\boldsymbol{n}$.
The determinant of a nilpotent matrix is 0. For example,

i) $A = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$ *is nilpotent because,* $\boldsymbol{A^2} = \begin{pmatrix} \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} \end{pmatrix} = \boldsymbol{O}$ *and*

ii) $B = \begin{pmatrix} 1 & 2 & 5 \\ 2 & 4 & 10 \\ -1 & -2 & -5 \end{pmatrix}$ *is nilpotent because,* $\boldsymbol{B^2} = \begin{pmatrix} \mathbf{0} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} \end{pmatrix} = \boldsymbol{O}$

iii) $C = \begin{pmatrix} 0 & 1 & 2 \\ 0 & 0 & -1 \\ 0 & 0 & 0 \end{pmatrix}$ *is nilpotent because,* $C^2 = \begin{pmatrix} 0 & 0 & -1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$ ***and*** $\boldsymbol{C^3} = \begin{pmatrix} \mathbf{0} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} \end{pmatrix} = \boldsymbol{O}$

Remark: In general, if $\boldsymbol{A}$ is an upper-triangular matrix (or lower triangular matrix) of order $n \times n$, and all its entries on the diagonal are zeros. Then we have, $\boldsymbol{A^n = O}$

Application of Invertible Matrices: Coding

Cryptology is the science of developing secret codes and using those codes for encrypting and decrypting data. Today, governments use sophisticated methods of coding and decoding messages.

There are many ways to **encrypt** a message. And the use of coding has become particularly significant in recent years (due to the explosion of the internet for example). One way to **encrypt** or **code** a message uses matrices and their inverse which is extremely difficult to break.

First, convert the message into a matrix B and consider a fixed invertible matrix A such that AB is possible to perform. Send the message AB. The receiver of the message AB decodes it using the inverse A^{-1} of the matrix A in order to **decrypt** or **decode** the message sent. Indeed, we have

$$A^{-1}(AB) = B$$

which is the original message.

This first matrix is A called the **encoding matrix** and its inverse A^{-1} is called the **decoding matrix.**

Keep in mind that whenever an undesired intruder finds A, we must be able to change it.

Remark on the type of encoding matrix A:

We should have a mechanical way of generating simple matrices A which are invertible and have simple inverse matrices. Note that, in general, the inverse of a matrix involves fractions which are not easy to send in an electronic form. The best is to have both A and its inverse with integers as their entries. In fact, we can use our previous knowledge to generate such class of matrices. Indeed, if A is a matrix such that its determinant is ± 1 and all its entries are integers, then A^{-1} will have entries which are integers. So how do we generate such class of matrices? One practical way is to start with an upper triangular matrix with ± 1 on the diagonal and integer-entries. Then we use the elementary row operations to change the matrix while keeping the determinant unchanged. Do not multiply rows with non-integers while doing elementary row operations. Let us illustrate this on an example.

How can we encode and decode a message??

To every letter we will associate a number. The easiest way to do that is to associate 0 to a blank or space, 1 to A, 2 to B, etc... as shown in table (1)

Another way is to associate 0 to a blank or space, 1 to A, -1 to B, 2 to C, -2 to D, etc... as shown in table (2)

__ = 0	**I** = 9	**R** = 18
A = 1	**J** = 10	**S** = 19
B = 2	**K** = 11	**T** = 20
C = 3	**L** = 12	**U** = 21
D = 4	**M** = 13	**V** = 22
E = 5	**N** = 14	**W** = 23
F = 6	**O** = 15	**X** = 24
G = 7	**P** = 16	**Y** = 25
H = 8	**Q** = 17	**Z** = 26

Table (1)

_ = 0	**I** = 5	**R** = - 9
A = 1	**J** = - 5	**S** = 10
B = - 1	**K** = 6	**T** = - 10
C = 2	**L** = - 6	**U** = 11
D = - 2	**M** = 7	**V** = - 11
E = 3	**N** = - 7	**W** = 12
F = - 3	**O** = 8	**X** = - 12
G = 4	**P** = - 8	**Y** = 13
H = - 4	**Q** = 9	**Z** = - 13

Table (2)

Example (1): Use table (1) to encode the message, **"PREPARE TO NEGOTIATE"** using the encoding matrix ,

$$A = \begin{pmatrix} -3 & -3 & -4 \\ 0 & 1 & 1 \\ 4 & 3 & 4 \end{pmatrix}$$

Solution

We use table(1) to assign a number for each letter of the alphabet. For simplicity, let us associate each letter with its position in the alphabet: A is 1, B is 2, and so on. Also, we assign 0 to a space between two words. Thus the message becomes:

P	R	E	P	A	R	E	-	T	O	-	N	E	G	O	T	I	A	T	E
16	18	5	16	1	18	5	0	20	15	0	14	5	7	15	20	9	1	20	5

Since we are using a 3 by 3 matrix, we break the enumerated message above into a matrix B of size 3 by 7

$$B = \begin{pmatrix} 16 & 16 & 5 & 15 & 5 & 20 & 20 \\ 18 & 1 & 0 & 0 & 7 & 9 & 5 \\ 5 & 18 & 20 & 14 & 15 & 1 & 0 \end{pmatrix}$$

We now encode the message by multiplying the above matrix $\boldsymbol{B}$ by the encoding matrix $\boldsymbol{A}$. This can be done by writing the above matrix and perform the matrix multiplication of that matrix $\boldsymbol{B}$ with the encoding matrix $\boldsymbol{A}$ as follows:

$$AB = \begin{pmatrix} -3 & -3 & -4 \\ 0 & 1 & 1 \\ 4 & 3 & 4 \end{pmatrix} \begin{pmatrix} 16 & 16 & 5 & 15 & 5 & 20 & 20 \\ 18 & 1 & 0 & 0 & 7 & 9 & 5 \\ 5 & 18 & 20 & 14 & 15 & 1 & 0 \end{pmatrix}$$

$$= \begin{pmatrix} -122 & -123 & -95 & 109 & -96 & -91 & -75 \\ 23 & 19 & 20 & 14 & 22 & 10 & 5 \\ 138 & 139 & 100 & 124 & 101 & 111 & 95 \end{pmatrix}$$

The columns of this matrix give the encoded message.

The encoded (encrypted) message is transmitted in the following linear form,

-122 , 23 , 138 , -123 , 19 , 139 , -95 , 20 , 100 , 109 ,14 , 124 , -96 , 22 , 101 , -91 , 10 , 111 , -75 , 5 , 95

Example (2): Decode the message sent in example (1) knowing that the sent message was encoded using the matrix,

$$\boldsymbol{A} = \begin{pmatrix} -3 & -3 & -4 \\ 0 & 1 & 1 \\ 4 & 3 & 4 \end{pmatrix}$$

Solution

To decode the sent message,

-122 , 23 , 138 , -123 , 19 , 139 , -95 , 20 , 100 , 109 ,14 , 124 , -96 , 22 , 101 , -91 , 10 , 111 , -75 , 5 , 95

The receiver writes this string (sent message) as a matrix $\boldsymbol{AB}$ of size $\mathbf{3} \times \mathbf{7}$ column matrices, then

$$\boldsymbol{AB} = \begin{pmatrix} -122 & -123 & -95 & 109 & -96 & -91 & -75 \\ 23 & 19 & 20 & 14 & 22 & 10 & 5 \\ 138 & 139 & 100 & 124 & 101 & 111 & 95 \end{pmatrix}$$

Thus, to decode the message, perform the matrix multiplication: $\boldsymbol{A^{-1}(AB) = B}.$

Therefore, we must find $\boldsymbol{A^{-1}}$ using the rule,

$$\boldsymbol{A^{-1}} = \frac{\mathbf{1}}{\boldsymbol{det(A)}}.\boldsymbol{adj(A)} = \begin{pmatrix} 1 & 0 & 1 \\ 4 & 4 & 3 \\ -4 & -3 & -3 \end{pmatrix}$$

Thus, to decode the message, perform the matrix multiplication

$$\boldsymbol{A^{-1}(AB)} = \begin{pmatrix} 1 & 0 & 1 \\ 4 & 4 & 3 \\ -4 & -3 & -3 \end{pmatrix} \begin{pmatrix} -122 & -123 & -95 & 109 & -96 & -91 & -75 \\ 23 & 19 & 20 & 14 & 22 & 10 & 5 \\ 138 & 139 & 100 & 124 & 101 & 111 & 95 \end{pmatrix}$$

$$\boldsymbol{B} = \begin{pmatrix} 16 & 16 & 5 & 15 & 5 & 20 & 20 \\ 18 & 1 & 0 & 0 & 7 & 9 & 5 \\ 5 & 18 & 20 & 14 & 15 & 1 & 0 \end{pmatrix}$$

The columns of this matrix, written in linear form, give the original message:

16 18 5 16 1 18 5 0 20 15 0 14 5 7 15 20 9 1 20 5

P R E P A R E - T O - N E G O T I A T E